

Institute of Actuaries of India

Subject CB2-Business Economics

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution: 1.5 Mark to each answer

1. D
2. A
3. A
4. A
5. C
6. C
7. C
8. B
9. B
10. D
11. C
12. D
13. D
14. B
15. D
16. B
17. B
18. C
19. B
20. C
21. B
22. D
23. C
24. A
25. B
26. A

Solution 27**i) Line 1 - Exports**

The *balance of trade in goods* is the net figure of exports and imports. We therefore have:

Exports = 425 - 40

Exports are therefore equal to **\$385m**

Line 6 - Balance of trade in services

The *balance of trade in services* is the net of exports and imports of services. We therefore have:

165 - 185 = *balance of trade in services*

The *balance of trade in services* is therefore **-\$20m**.

Line 8 - Income payments

The *net income receipts* (line 9) are the net of *income receipts* and *income payments*. We therefore have:

190 - income payments = 15

Income payments are therefore **-\$175m**.

Line 11 - Net income flows

Net income flows are the net of net *income receipts* and *net current transfers*. We therefore have:

15 - 50 = net income flows

Net income flows are therefore **-\$35m**.

Line 12 - Current account balance

The *current account balance* is the sum of the *balance of trade in goods* (line 3), the *balance of trade in services* (line 6) and *net income flows* (line 11). We therefore have:

-40 - 20 - 35 = current account balance

The *current account balance* is therefore **-\$95m**.

Line 18 - Capital and financial account balance

The balance of payments has to balance. In other words the net of the *current account balance* and the *capital and financial account balance* must be 0.

Since the current account is **-\$95**, the *capital and financial account balance* must be **+\$95m**.

[3]

ii)

S.No.	Transaction	Credit	Debit
a.	Japanese purchase of Indian T bonds	credit	
	Japanese payment using Mumbai bank		debit
b.	Indian citizen having a meal in Paris		debit
	Paying the meal with American Express	credit	
c.	Export of programming service	credit	
	British payment out of its account in India		debit

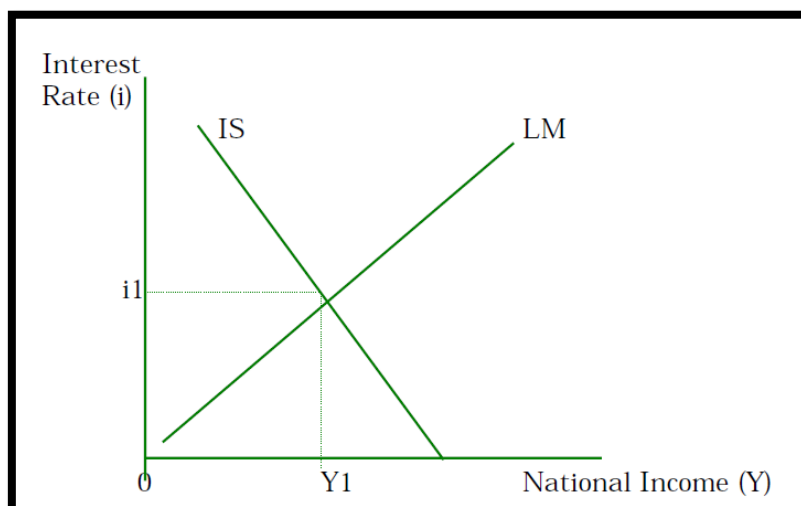
[2.5]

[5.5 Marks]

Solution 28:

i) Points on the IS curve show combinations of interest rates and national income at which injections equal withdrawals and consequently the total output (national income) equals total expenditure, giving equilibrium in the markets for goods and services. [1]

Points on the LM curve show combinations of interest rates and national income at which the demand for money (liquidity) equals the supply of money, giving equilibrium in the money markets. [1]



[1]

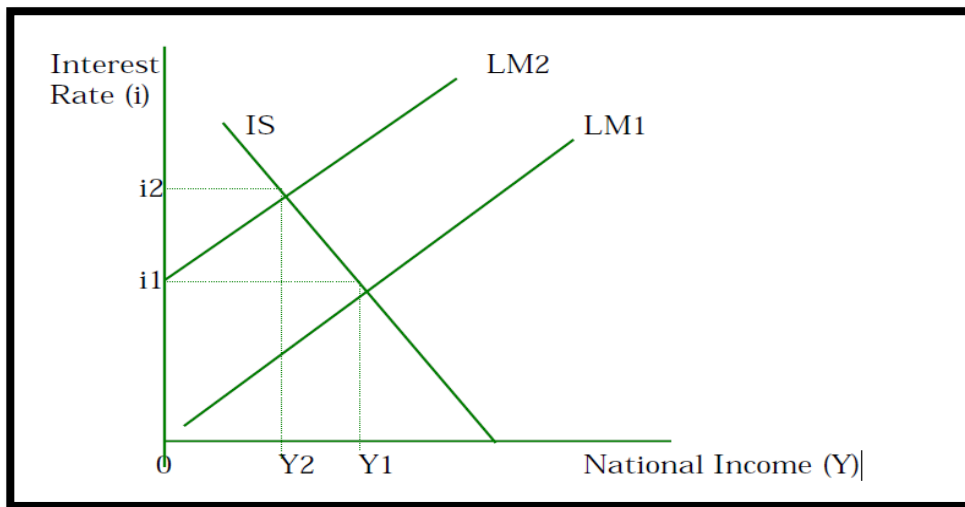
The IS-LM curves are drawn on a graph with interest rates on the vertical axis and national income on the horizontal axis. The IS curve slopes downward from left to right showing that lower rates of interest lead to higher levels of national income. The LM curve slopes upward from left to right showing that in order to obtain equilibrium in the money market, given the supply of money, higher levels of national income are associated with higher rates of interest. [1]

At the rate of interest and level of national income where the IS-LM curves intersect in the graph both the money market and the market for goods and services are in equilibrium. [1]

[5]

- ii) An increase in the average price level will result in a reduction in the real value of the money supply. A reduction in the money supply will shift the LM curve to the left giving an increase in the equilibrium rate of interest and a fall in the equilibrium level of national income. [1]

The reduction in the money supply gives an excess demand for money causing rates of interest to rise. The rise in interest rates discourages investment and consumer expenditure, reducing injections and causing the equilibrium level of national income to fall. [1]



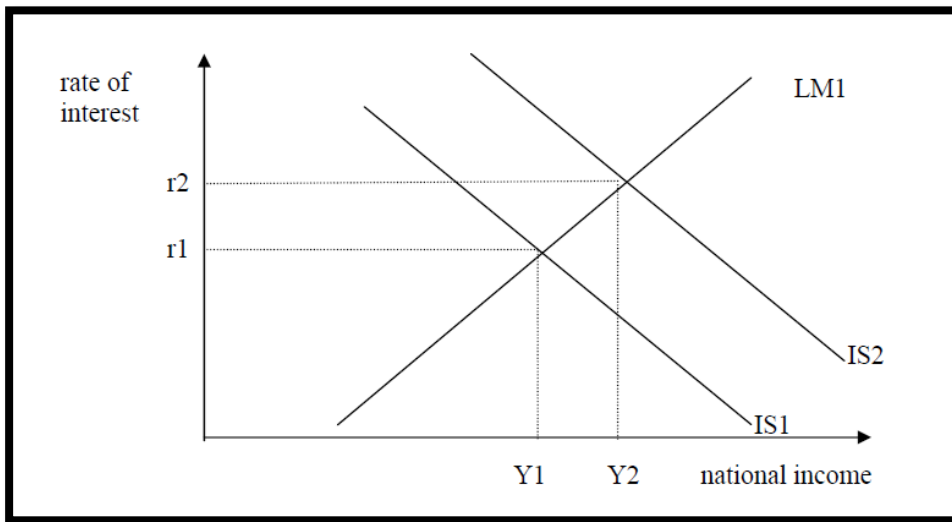
[1]

[3]

- iii) With an expansionary fiscal policy financed by the sale of bonds, the government sells bonds and uses the proceeds raised to finance increased government expenditure. The effect on the money supply should be zero as the money raised from the bond sales is spent by the government. [1]

However, the increase in the supply of bonds for sale pushes down bond prices and raises the interest rate. There is a multiplier effect associated with the increase in government expenditure which is only partially offset by the effect of rising interest rates on consumption and investment so there is a net increase in the level of national income. [1]

This is depicted in the diagram below with the increase in government expenditure shifting the IS curve to the right from IS_1 to IS_2 and the interest rate rising from r_1 to r_2 and the level of national income rising from Y_1 to Y_2 .



[1]

[3]

[11 Marks]

Solution 29:**i) Hysteresis**

Persistence of unemployment even when the deficiency of aggregate demand that caused it, no longer exists. [1]

For e.g., a recession causes the long term unemployed to lose their skills and motivation, hence they might not easily obtain work when the economy begins to recover. [1]

ii) Structural Deficit is the public sector deficit that would occur if the economy were operating at the potential level of national income or output, i.e., where there is no excess or deficiency of aggregate demand and therefore a zero output gap. [1.5]

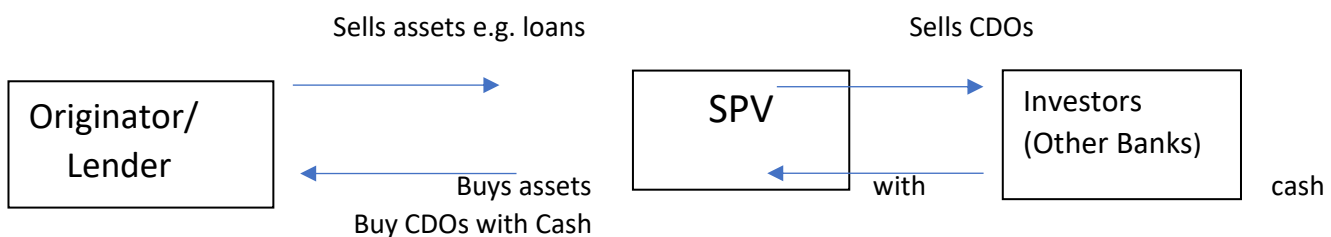
iii) Accelerator Coefficient

The accelerator theory states that investment is determined by the rate of change of national income. If national income rises, then the level of capital needed to produce more goods and services also rises. This is called induced investment. [1]

The amount of induced investment depends on the change in national income, i.e., $I = \alpha \Delta Y$, where, α , the accelerator coefficient, is the amount by which induced investment depends on national income. [1]

iv) Securitisation with diagram

Securitisation is the process of pooling assets, such as loans or mortgages, into marketable securities, such as bonds, backed by these assets. [1]



[1]

v) Repo rate is the rate at which the Reserve Bank of India (RBI) grants the loan to the commercial banks against government securities. [0.5]

For controlling the inflation, RBI uses this tool whereby the rate is increased to reduce the commercial bank's borrowings, which in return pulls down the inflation. [0.5]

Reverse repo rate is exactly opposite to a Repo rate; it is an interest rate at which the commercial bank grants the loan to the Central Bank of India i.e. RBI. [0.5]

This is a monetary tool used by RBI to control the money supply in the country, i.e. with the increase in the rate, the flow of money in the economy decreases as the banks will now invest its money with RBI due to higher interest rates. [0.5]

The Reverse repo rate is always lower than a repo rate. [0.5]

[10 Marks]

Solution 30:

- i) a. Money supply will be unaffected because the public's cash is withdrawn from banks and government spends it, so it is redeposited in the banking system. Hence, no overall change in the amount of cash held in the banking system. [1]
- b. Increase because there will be more cash deposited in banks, rather than being held outside of the banking system, which they can use to create credit. [1]
- ii) a. A fall in the national income leads to a decrease in the transactions demand for money. The demand for money curve will shift to the left. [1]
- b. A decrease in the expected value of domestic currency will cause a decrease in the demand for the currency, i.e., the demand for money curve will shift to the left. [1]

[4 Marks]

Solution 31:

The Government should allow maximum commission regulation to continue:

Negative aspect of regulation:

- The imposition of maximum level of commission reduces the ability of insurers to compete on the commission terms they offer to intermediaries.
- Intermediaries may consider the terms of regulation unfair as it reduces the maximum amount of income they can receive for each sale.

Positive aspects of regulation:

- This regulation stops insurers from raising commissions to intermediaries as a marketing tool to sell more policies.
- In absence of such arrangement, there is a danger that intermediaries will be tempted to sell contract with highest commission rather than the contract most appropriate to the policyholders.
- Consumers benefit from the cap of the amount of commission that must be loaded into their premiums.
- Commission rates are not fixed. Intermediaries like brokers are free to compete by offering to take less commission than maximum. [3 Marks]

Solution 32:

- i) Perfectly contestable markets:
- a. A contestable market occurs when there is freedom of entry and exit into the market.
 - b. In a contestable market there will be low sunk costs.
 - c. Due to freedom of entry and exit, the existing firm always faces the threat of new firms entering the market.
 - d. This threat to entry will keep the prices competitive and profits low.

Contestable markets and consumer interest:

Contestable markets bring the following benefits of competitive markets:

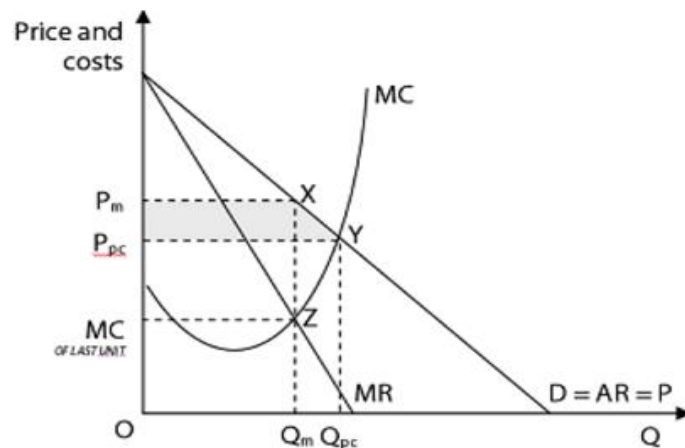
1. Lower prices- allocative efficiency
2. Increased incentives for firms to cut cost through economies of scale
3. Increased incentives for firms to meet consumers' interests
4. Potential competition will keep profits low

ii) Example-

1. Airline routes-

- a) Airline travel is contestable as there are existing firms in the market.
- b) Existing airline would have recovered sunk cost at entry, therefore will experience low entry costs for other routes
- c) The new airlines entering the airline industry may experience higher sunk costs and find it difficult to transfer capital outlays to another use.
- d) If the existing airline enters another route and proves to be unsuccessful, they can withdraw from that route at relatively low cost.

iii) Diagram for perfectly contestable market-



[1]

[6 Marks]

Solution 33:

i) Negative Externality Definition -

Negative externalities occur when production and/or consumption impose external costs on third parties outside of the market for which no appropriate compensation is paid. This causes social costs to exceed private costs.

An example of a negative externality:

- i. People may smoke in public places and ignore harmful impact of toxic passive smoking on non-smokers.
- ii. The passive smoking may harm health of non-smoker and may cause various respiratory diseases.
- iii. In young ladies passive smoking may cause pre-mature child birth and asthma.
- iv. Passive smoking increases the risk of respiratory illnesses in children, including asthma, bronchitis and pneumonia.
- v. Healthcare expenditure involved in treatment of smoking related illnesses

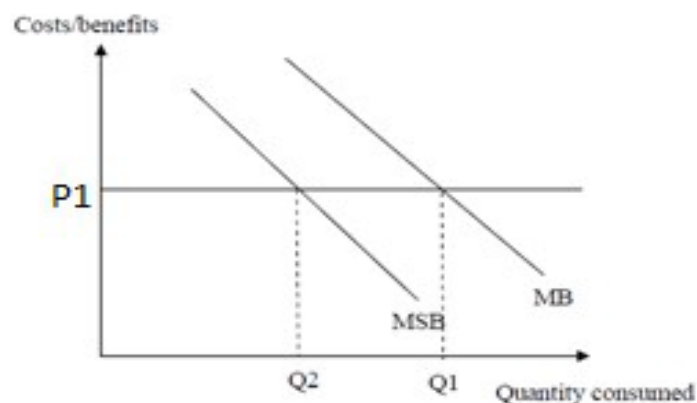
ii) Price Related Intervention:

- a) The Government imposes a tax on the producer that reflects the marginal cost of the externality.
- b) Fines are imposed if smoking is done in public places

Non- Price Related Intervention:

- a) Smoking is banned in public places like gardens, restaurants, libraries, work places, etc.
- b) A law prohibiting any manufacturer, retailer, or distributor of tobacco from selling or distributing products to persons under age eighteen.
- c) Use of media / advertisements, public education to emphasize ill effects of smoking and motivate smokers to leave smoking
- d) Specially designed medical care and or programs for smokers to quit smoking.

- iii) Diagram- marginal social benefit curve and marginal private benefit curve when a negative externality in consumption exists-



Q1- Consumption by private sector

Q2- Socially optimal level of consumption

MSB- Marginal Social Benefit

MB- Marginal Private Benefit

P1- Market price

[8 Marks]

Solution 34:

- i) Difference between actual and potential growth-

The term actual economic growth refers to increases in the real Gross Domestic Product or real Gross National Product. The economic growth rate measures the percentage increase in the real Gross Domestic Product from one year to the next. The term actual economic growth refers to the increase in the trend rate of real Gross Domestic Product rather than the actual increase in output.

Potential economic growth refers to the speed at which the economy could grow and so it measures the percentage increase in an economy's capacity to produce.

- ii) Main factors that explain potential growth

The quantity of resources:

- capital
- labour
- And raw materials.

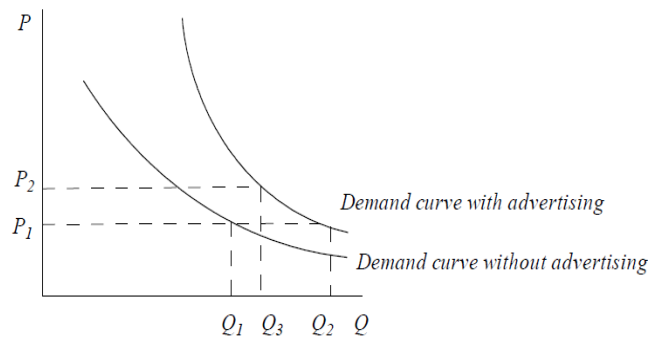
The productivity of resources, which depends on:

- technological progress
- education, training and healthcare

[4.5 Marks]

Solution 35:

Diagram showing intended effect of advertising:



P1- Original price of product

Q1- Original quantity demanded

Q2- Quantity demanded after advertisement effect sold at original price of product

P2- Increase in price after advertising

Q3- Quantity demanded at price P2

- Advertising a product is intended to shift the product's demand curve to the right
- to make the product's demand less price-elastic
- At price P_1 , the quantity demanded is Q_1 . After advertising, the demand curve shifts to the right enabling an increased quantity (Q_2) to be sold at the original price
- However, the firm may be able to make greater profits than this by increasing its price to P_2 and still increasing sales (to Q_3).

How advertising achieves these effects:

- Advertising can shift the demand curve to the right by bringing the product more to Consumers' attention
- Advertisement informs consumers about the product's attractive features, especially relative to those of competitors' products.
- Advertising can make the product's demand less price-elastic by creating brand loyalty and so reducing the number of perceived substitute goods.

[5 Marks]

Solution 36:

A risk neutral agent does not require an increased expected return to take on increased risk. If people are risk neutral, they will always choose the option with the highest expected return.

A risk adverse agent requires an increase in expected return to take increased risk. If people are risk averse they will never accept a gamble if it has the same expected return as the pay-off from not taking a gamble.

[2 Marks]

Solution 37:

There are two key requirements for adverse selection to be a problem:

- Some people know that they are relatively bad risks
- The insurance company is not able to tell who is a bad risk and who is a good risk.

Steps taken by insurer:

- Insurance companies screen potential policyholders to try to find out lots of information about them.
- Policyholders can then be put in small, reasonably homogeneous pools and charged appropriate premiums.

[2 Marks]
